Bhavadeep (Bob) Kallepalle

Senior Software Engineer | AI & Agentic Platforms

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PROFESSIONAL SUMMARY

Senior Software Engineer with 10+ years across platform engineering, application development, and IoT, now focused on AI and agentic systems. Lead an engineering team and own sourcing infrastructure supporting budget planning and cost of goods calculations. Build reusable agent platforms and manufacturing automation across diagnostics, factory planning, and supply chain workflows.

TECHNICAL SKILLS

Languages & frameworks: C#, .NET, Python, TypeScript, Java, SQL; ASP.NET Core, Node.js, WPF, Blazor.

AI & agents: Microsoft Agent Framework, Semantic Kernel, Azure OpenAI / AI Foundry, MCP, RAG, tool calling, local models, model routing, evaluation harnesses.

Cloud & data: Azure Functions, Container Apps, App Insights, Storage, Event Grid, Service Bus; PostgreSQL, Cosmos DB, Kusto, Redis.

Architecture & delivery: Distributed systems, microservices, event-driven design, CQRS, HAL; Azure DevOps, CI/CD, Bicep/IaC, TDD, mentoring.

EXPERIENCE

Senior Software Engineer | Microsoft

Redmond, WA | Sep 2025 - Present

- Lead an engineering team owning sourcing infrastructure critical to budget planning and cost of goods calculations; set technical direction and delivery priorities.
- Drive AI-powered manufacturing diagnostics, test execution, and telemetry analysis, alongside factory planning and supply chain automation.
- Architect reusable agent capabilities: tool and skill contracts, grounded retrieval, orchestration, human-in-the-loop review, and evaluation/telemetry pipelines.
- Define AI adoption patterns and guardrails; mentor engineers on agent design, evaluation, and responsible AI.

Software Engineer II | Microsoft

Redmond, WA | Sep 2021 - Sep 2025

- Owned a modular Windows service framework for Surface and Xbox manufacturing tests, with dynamically loaded modules and ASP.NET service endpoints.
- Built hardware abstraction that decoupled device types and vendors, enabling new hardware onboarding without framework changes.
- Delivered scalable RF test solutions for Surface and pioneered AI architecture across manufacturing workflows.

Software Development Engineer | Microsoft

Redmond, WA | Jun 2020 - Sep 2021

- Deployed a Windows 10 DHCP server service across hundreds of industrial PCs on manufacturing lines.
- Integrated a customer-zero program into the Manufacturing Test Platform; delivered GPU, touchpad, and sensor tests.

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SELECTED PROJECTS

Hydra Edge | Privacy-Aware Local Model Routing for VS Code

MAIDAP × OCTO Summer Hackathon, Microsoft | 2026

- Co-built privacy-aware routing that directs sensitive AI requests to on-device models; owned VS Code integration, editor/file context, and routing interfaces.
- Designed secret, PII, and sensitive-code classification using rule matching, entropy analysis, regex/Luhn checks, and path/diff heuristics; authored architecture and design artifacts.
- Placed within one point of the winning entry; recognized by engineering leadership and being ported for Surface RTX Spark by Windows + Devices.

Agent Workflow Engine | Visual, Observable Multi-Model Workflows

2026

- Co-built reusable AI workflows and a visual editor for configuring steps, flowchart views, validation, and save/load.
- Connected execution to live step states, failure diagnostics, usage, and timing; authored implementation, critique, and repair flows with budgets and verification gates.
- Compared model configurations for correctness, completion, cost, and runtime; improved failure reporting and Windows reliability.

EARLIER EXPERIENCE

Software Development Engineer II | CalAmp

Irvine, CA | Jan 2020 - Jun 2020

- Coordinated engineering teams, vendors, and partners on wireless telematics solutions.

Android Platform Engineer | CalAmp

Carlsbad, CA | Feb 2018 - Jan 2020

- Built Android tablet/IoT platforms and vehicle-data SDKs; delivered dash camera support and a custom IDE extension.

Android Developer | ClassCalc

Greater Los Angeles Area | Jul 2017 - Jan 2018

- Built a backward-compatible scientific and graphing calculator UI; coordinated iOS/Android consistency and code quality.

Software Engineer | QuEST Global

Trivandrum, India | May 2014 - Jul 2015

- Developed and tested Toshiba C-ARM healthcare software; built a multithreaded log analyzer and ported MATLAB image-processing algorithms to CUDA.

EDUCATION

M.S. Computer Science | California State University, Long Beach | 2015 - 2017

B.Tech. Electrical & Electronics Engineering | National Institute of Technology, India | 2010 - 2014

PATENT PUBLICATION

Co-inventor: Technologies for driver behavior assessment

WO2021168387A1 | WIPO/PCT application publication | CalAmp Corp | Aug 26, 2021

Combines vehicle sensor data and event-triggered video capture to assess driver behavior.